



Jiarui Liang 梁家睿

ljr23@mails.tsinghua.edu.cn

Website: <https://liang-jiarui.github.io/>



Education

- 2023.09 - Tsinghua University, PhD candidate, Supervisor: Associate Professor Bo Zheng
- GPA: 4.0/4.0
- 2019.09 - 2023.06 China University of Petroleum (Beijing), Environment Engineering
- GPA: 4.18/5.0

Research

- 2025.08- **Evolution of wildfire emissions and associated Earth system impacts**
- Based on the Community Earth System Model (CESM), this study quantifies the atmospheric environmental impacts and climatic effects of global wildfire activity emissions.
- 2023.09- **Anthropogenic nitrous oxide emissions in China: trends, variations and driving factors**
- Using emission inventory and machine learning to quantify China's anthropogenic nitrous oxide emissions, and explore mitigation pathways compatible with the 1.5 °C target
- 2024.12-2025.08 **Source apportionment of summer ozone in Beijing based on a machine learning-enhanced atmospheric chemical transport model framework**
- Build a machine learning-enhanced atmospheric chemical transport model framework (GEOS-Chem and WRF-CAMx), quantify summer ozone sources in Beijing using the Ozone Source Apportionment Technology (OSAT), and complete an English academic paper
- 2022.07-2025.02 **Source-sink budgets and spatiotemporal pattern of carbon dioxide, methane and nitrous oxide**
- Based on global atmospheric inversion results of greenhouse gases from the global carbon project, conduct ensemble analysis of the source-sink budgets and spatiotemporal distribution of greenhouse gases in china
- 2021.09-2023.09 **Operational optimization of industrial circulating water systems**
- Establish a mixed-integer nonlinear optimization model for the circulating water system, optimize converter frequency and valve opening, and solve the model based on the genetic algorithm
 - publish two english academic papers as the first author

Publications

1. Liang, J[#], Qiu, J[#], Zheng, B^{*}, et al. Transboundary air pollution masks a decade of ozone mitigation in Chinese cities. (Prepareing for submission)
2. Liang, J., Li, Y., Wang Y^{*}, et al. Long-term operation optimization of circulating cooling water systems under fouling conditions, Chinese Journal of Chemical Engineering, 2024.
3. Liang, J., Li, L., Wang Y^{*}, et al. Operation optimization of existing industrial circulating water system considering variable frequency drive, Chemical Engineering Research and Design, 2022.
4. Gu, C., Liu, S., Liang, J., et al. Degradation of 1,2,3-trichloropropane by pyrite activating sodium percarbonate and the implications for groundwater remediation, Journal of Environmental Chemical Engineering, 2023.

Research Skills

1. Proficient in Linux operating system and High Performance Computing (HPC) clusters; skilled in Python programming, with a grasp of applications of conventional Machine Learning and Deep Learning models
2. Experienced in meteorological data processing, compilation and execution of global and regional Earth System Models and Atmospheric Chemical Transport Models (CESM, GEOS-Chem, WRF, CAMx, CAM-Chem)